

Position

+ Position(float px, float py, bool forMovableObject=true, float detectabilityRadius=game::DEFAULT_DETECTABILITY_RADIUS)
+ virtual ~Position()
+ int getSpaceWidth() const
+ int getSpaceHeight() const
+ float getX() const
+ float getY() const
+ float getDistanceWith(Position const &autre) const
+ float getDistanceWith(float const &x, float const &y) const
+ void setPosition(float posX, float posY)
+ MyVector getPosition() const
+ virtual void setPositionMovementLimit(sf::IntRect const &hall)
+ void operator+=(MyVector const &v)
+ void operator=(Position const &p)
+ static void initSpace(int width, int height)

MyVector

+ float x
+ float y

+ void operator+=(MyVector const &v)
+ void operator-=(MyVector const &v)
+ bool operator==(MyVector const &v)
+ bool operator!=(sf::Vector2f const &v)
+ MyVector operator*(float coefficient) const
+ sf::Vector2f toVector2f()
+ static MyVector createFromAngle(float simpleSize, float angleInDegree)

GameObject

const int m_id
std::string m_gameObjectName
sf::Texture m_texture
sf::Sprite m_body
static std::map< std::string, GameObject * > gameObjects
static int m_count
static float m_time

+ GameObject(std::string _view const &imageSpritePath)
+ virtual ~GameObject()
+ virtual void show(sf::RenderTarget &window)
+ virtual void update()=0
+ int getGameObjectId() const
+ const Position * getPosition() const
+ Position * getNonConstPosition()
+ std::string getGameObjectName() const
+ void setGameObjectName(std::string name)
+ virtual void onCollisionEnter(Collision const &collision) const =0
+ sf::Sprite const & getSprite() const
+ bool isArigidBody() const
+ void fromColliderToRigidBody() const
+ Collider const & getCollider() const
+ static void SetTime(float t)
+ static std::map< std::string, GameObject * > getGameobjects()
void init()
virtual void initPosition(float posX, float posY)=0
static void incrementCount()

Collider

std::string m_colliderType
static GameObjectMap m_objects

+ Collider(GameObject const &parent, std::string typeOfCollider)
+ virtual ~Collider()
+ GameObject const & getParent() const
+ virtual std::vector< sf::Vector2f > getContactPoints() const =0
+ virtual bool touchEachOther(Collider const &collider) const =0
+ static void update()
+ static void removeObject(int gameId)

Rigidbody

+ Rigidbody(GameObject &parent)
+ ~Rigidbody()
+ void onAnotherRigidBodyDetection()

#m_position

#m_speed

#parent

#m_collider

#m_parent

#m_rigidbody